

Lab Session

Maximize $f_1(\mathbf{x}) = x_1$, $f_2(\mathbf{x}) = x_2$,
subject to $x_1 + x_2 \leq 1$, $0 \leq x_1$, $0 \leq x_2$.

Pareto front is a line segment between $(0, 1)$ and $(1, 0)$.

Task 1: When the reference point is $(0, 0)$, show a single solution which maximizes the hypervolume indicator.

Task 2: When the reference point is $(0, -1)$, show a single solution which maximizes the hypervolume indicator.

Task 3: When the reference point is $(0, 0)$, show a set of two solutions which maximizes the hypervolume indicator.

Task 4: When the reference point is $(-2, -2)$, show a set of two solutions which maximizes the hypervolume indicator.

Task 5: When the reference point is $(-2, -2)$, show a set of five solutions which maximizes the hypervolume indicator.

Task 6: When an infinitely large number of reference points are uniformly given over the entire Pareto front, show a set of five solutions which minimize the IGD indicator.